SHEIKH SAIMON AHAMED ABIR

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EDUCATION

M.Sc. in Energy and Environmental Technology — **Aug 2021 – May 2023**
Department of Process, Energy and Environmental Technology
University of South Eastern Norway
Grade Scale: C

B.Sc. in Mechanical Engineering — **May 2015 – April 2019**
Department of Mechanical and Production Engineering
Ahsanullah University of Science and Technology
CGPA: 3.202/4.00

ACADEMIC INTERESTS

- Renewable Energy
- Wind Energy
- Circular Economy

RESEARCH AND PROJECT EXPERIENCE

Master's Thesis

Development of Circular Economy Routes for Refractory Materials Waste

Supervisor: Prof. Dr. Chandana Ratnayake, University of South Eastern Norway

- ✓ Prepared mixed (500 g) refractory waste samples from steel and cement industries representing industrial conditions across multiple size fractions
- ✓ Performed particle classification using a cross-flow air classifier at different air velocities (8, 12, 14 m/s)
- ✓ Conducted detailed particle size distribution (PSD) analysis using multi-stage sieve towers
- ✓ Generated PSD and Tromp curves to evaluate separation behavior and classifier performance
- ✓ Calculated key performance parameters including cut size (d50), sharpness of separation, and imperfection factor
- ✓ Compared results across different operating conditions and particle size ranges to assess classification efficiency

Master's Project(s)

Bunkering of Hydrogen Fuel onto Maritime Vessels

Course: Hydrogen and Energy Technology

- ✓ Reviewed literature and gathered information related to hydrogen fuel bunkering risks and safety
- ✓ Developed design of optimized bunkering system of hydrogen fuel from fuel station to maritime vessel

Potentiality of Biogas Production from Organic Food Waste

Course: Water Treatment and Environmental Bio-Technology

- ✓ Prepared feed samples for Bioreactor testing
- ✓ Measured pH and methane gas generation rate from the samples
- ✓ Performed Gas Chromatography (GC) analysis to ensure the presence of Biogas

Energy Management Systems Review

Course: Masters Project

- ✓ Reviewed research papers on BEMS design, strategies, and energy-saving performance
- ✓ Analyzed standards and regulations (ISO 50001, EU directives) to establish framework
- ✓ Studied EMS architecture including sensors, IoT communication, data processing, and control systems
- ✓ Conducted comparative evaluation of commercial and open-source EMS platforms
- ✓ Developed comprehensive understanding of BEMS operation and their role in improving energy efficiency and sustainability

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Bachelor's Thesis

Design of a 3kW Horizontal Axis Wind Turbine for Prospective Locations of Bangladesh

Supervisor: Prof. Dr. Mazharul Islam, Ahsanullah University of Science and Technology

- ✓ Designed and simulated a 3kW Horizontal Axis Wind Turbine using QBlade
- ✓ Worked with (NREL S-series and NACA series) for blade design and performance testing
- ✓ Validated results using reference experimental Cp–TSR data
- ✓ Conducted a wind resource assessment of Bangladesh by numerical analysis using Weibull distribution function method
- ✓ Calculated annual energy production using Weibull parameters and turbine simulation

PUBLICATIONS(s) AND RESEARCH

(1) S.S.A. Abir and M.R. Hossain, "**Strategic Roadmap for Offshore Wind Energy Development in Bangladesh: A Review**" Oxford Open Energy (Published); Publisher: Oxford University Press;
DOI: https://doi.org/10.1093/ooenergy/oiaf015

(2) Abir, Sheikh Saimon Ahamed, et al. "**Machine-Learning-Accelerated Design Exploration of Floating Offshore Wind Turbine Mooring Systems Using OpenFAST-Based Parametric Simulations**" Results in Engineering (Under Peer Review); Publisher: Elsevier; SSRN DOI: https://dx.doi.org/10.2139/ssrn.6442479

(3) Abir et al. "**High Hub-Height Wind Resource Assessment for Coastal Bangladesh: Energy-Based Validation and Turbine Height Optimization Under Cost Scaling**" Renewable Energy & Power Quality Journal (RE&PQJ) (Under Peer Review); Publisher: Cultech Publishing Malaysia.

STANDARDIZED TEST SCORES

IELTS	Score: 7.5, CEFR: C1	(L: 8.5, R: 7.0, W: 6.5, S: 7.0)

EXPERTISE

Programming	: Python	**Data Analysis Tool**	: MS Excel
CAD	: SolidWorks, Inventor, AutoCAD	**Simulation Tool**	: OpenFast, QBlade
Documentation	: Latex, MS Word		

PROFESSIONAL EXPERIENCE

Mechanical Engineer **Oct 2019 – Oct 2020**

Saleh Steel Industries Limited, Dhaka, Bangladesh

- Created 2D utility piping layouts using AutoCAD ensuring efficient installation
- Prepared fabrication drawings for structures, supports and flanges
- Prepared BOQ generated from AutoCAD using Excel and macro
- Procured raw materials and manpower for construction work

Service Engineer **April 2025 – Present**

Safe zone High-Tech and Logistics Limited, Dhaka, Bangladesh

- Performed installation, testing, and commissioning of fire detection, alarm, and protection systems including sprinklers, hydrants, and extinguishers
- Assisted in site surveys and system design implementation according to project specifications and safety standards (NFPA and local codes)
- Prepared technical reports, maintenance logs, and service documentation for completed tasks and inspections
- Coordinated with senior engineers and project teams during installation and commissioning activities at industrial and commercial sites